

Seminar 12

Logici și Inele Boole

Def: Latticea $(A, \vee, \wedge)$ este o lattice Boole dacă:

1) A este distributivă

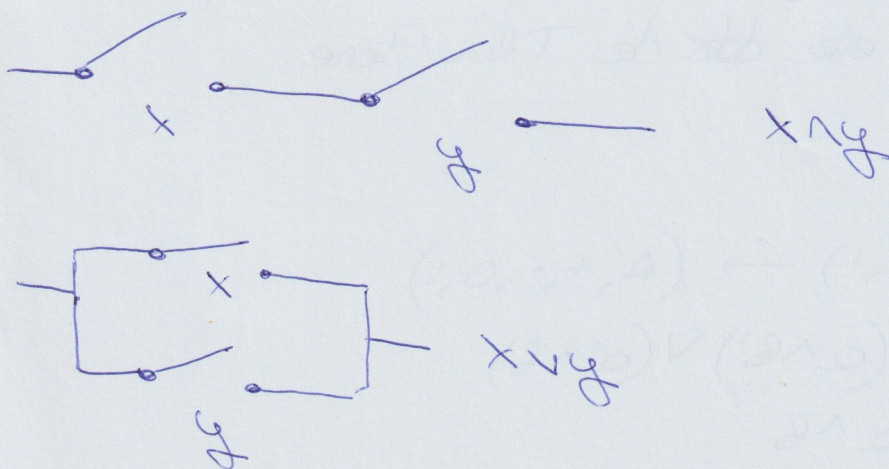
2) $\exists$ cel mai mic element $0 = \min A$

3) $\exists$ cel mai mare element $1 = \max A$

4) $\forall a \in A, \exists a' \in A$ a.t. $\begin{cases} \inf\{a, a'\} = 0 \\ \sup\{a, a'\} = 1 \end{cases} \Leftrightarrow \begin{cases} a \wedge a' = 0 \\ a \vee a' = 1 \end{cases}$

Notatie $(A, \vee, \wedge, 0, 1, ')$
complementa

Aplicații: - Electronica



- Simplificarea problemelor

Def: Inelul asociativ cu unitate $(A, +, \cdot, 0, 1)$ s.m. Inel Boole dacă $\forall x \in A: x^2 = x$

Obs: În particular se poate scrie: $1+1=0$

$$(1+1)^2 \stackrel{\text{def}}{=} 1+1 \Leftrightarrow \cancel{1+1+1+1} (1+1)(1+1) = 1+1 \Leftrightarrow$$

$$\Leftrightarrow 1+1+x+x = x+x \Rightarrow 1+1=0 \quad | \cdot x$$

$$\Rightarrow x+x=0 \quad \forall x \in A \Rightarrow \boxed{x+x=0} \quad | \cdot 1$$

$$\Rightarrow 2x = 0$$

$$3x = x$$

$$4x = 0$$

Obs 2) Orice inel Boole este comutativ

$$(x+y)^2 = x+y \rightarrow \neq x^2 + 2xy + y^2 \text{ nu stam ca } 2xy = 0$$

$$\Leftrightarrow (x+y)(x+y) = x+y$$

$$\Leftrightarrow x^2 + xy + yx + y^2 = x+y$$

$$\Leftrightarrow x + xy + yx + y = x+y$$

$$\Rightarrow xy + yx = 0 \quad | +yx$$

$$xy = yx \quad \forall x, y \in A$$

Obs: Cele 2 notiuni (latrice Boole si inel Boole) sunt echivalente, iar legatura este data de Teoria Stone.

TR Stone:

$$a) (A, \vee, \wedge, 0, 1, ') \rightarrow (A, +, \cdot, 0, 1)$$

$$\begin{cases} a+b = (a \wedge b') \vee (a' \wedge b) \\ a \cdot b = a \wedge b \end{cases}$$

$$b) (A, +, \cdot, 0, 1) \rightarrow (A, \vee, \wedge, 0, 1, ')$$

$$\begin{cases} a \vee b = a + b + a \cdot b \\ a \wedge b = a \cdot b \\ a' = 1 + a \end{cases}$$

Dem: a) Stăm că $(A, \vee, \wedge, 0, 1, ')$ latrice Boole.

Dem că $(A, +, \cdot, 0, 1)$ este un inel Boole

Σ Dem of $(A, +)$ group axiom

For operation ~~etc~~ binary-def

Associative: For $a, b, c \in A$

$$\text{Dem of } \underbrace{a + (b + c)}_{M_0} = \underbrace{(a + b) + c}_{M_1}$$

$$\begin{aligned} M_0 \Rightarrow (a \wedge (b+c)') \vee (a' \wedge (b+c)) &= (a \wedge ((b \wedge c') \vee (b' \wedge c))') \vee \\ &\vee (a' \wedge (b \wedge c') \vee (b' \wedge c)) = \end{aligned}$$

$$\stackrel{\text{distr}}{=} (a \wedge ((b \wedge c') \vee (b' \wedge c))') \vee (a' \wedge b \wedge c') \vee (a' \wedge b' \wedge c) = \textcircled{*}$$

$$\begin{aligned} M_1 \Rightarrow ((a \wedge b) \wedge c') \vee ((a \wedge b)' \wedge c) &= \\ = (((a \wedge b') \vee (a' \wedge b)) \wedge c') \vee (((a \wedge b') \vee (a' \wedge b))' \wedge c) &= \\ \stackrel{\text{distr}}{=} \underbrace{(a \wedge b' \wedge c')} \vee \underbrace{(a' \wedge b \wedge c')} \vee \underbrace{(((a \wedge b') \vee (a' \wedge b))' \wedge c)} \end{aligned}$$

$$\begin{aligned} \textcircled{*} \stackrel{\text{De Morgan}}{\Rightarrow} (a \wedge ((b \wedge c')' \wedge (b' \wedge c)')) \vee (a' \wedge b \wedge c') \vee (a' \wedge b' \wedge c) &= \\ = (a \wedge ((b' \vee c) \wedge (b \vee c'))) \vee (a' \wedge b \wedge c') \vee (a' \wedge b' \wedge c) &= \end{aligned}$$

$$\begin{aligned} \stackrel{\text{distr}}{=} (a \wedge (\cancel{b \wedge b'} \vee (b' \wedge c') \vee (c \wedge b) \vee \cancel{c \wedge c'})) \vee \\ \text{over} \quad \vee (a' \wedge b \wedge c') \vee (a' \wedge b' \wedge c) = \end{aligned}$$

$$\begin{aligned} \stackrel{\text{distr}}{=} \underbrace{(a \wedge b' \wedge c')} \vee \underbrace{(a \wedge c \wedge b)} \vee \underbrace{(a' \wedge b' \wedge c')} \vee \underbrace{(a' \wedge b \wedge c)} \end{aligned}$$

Commutativité $a+b = b+a$

$$\begin{aligned} a+b &= (a \wedge b') \vee (a' \wedge b) \stackrel{\text{Com}}{=} (a' \wedge b) \vee (a \wedge b') = \\ &\stackrel{\text{Com}}{=} (b \wedge a') \vee (b' \wedge a) = b+a \end{aligned}$$

Elem neutre $a+0 \stackrel{?}{=} a$

$$a+0 = (a \wedge 0') \vee (a' \wedge 0) = (a \wedge 1) \vee 0 = a \vee 0 = a$$

Elem opuse $a+a \stackrel{?}{=} 0$
($-a=a$)

$$\begin{aligned} a+a &= (a \wedge a') \vee (a' \wedge a) \\ &\stackrel{\text{Com}}{=} (a \wedge a') \vee (a \wedge a') \\ &= a \wedge a' \\ &= 0 \end{aligned}$$

II $(A, \cdot)$ monoid

$$a \cdot b = a \wedge b \quad \text{associativité} \quad \swarrow \text{max } A$$

$$\text{Elem neutre : } a \cdot 1 = a \wedge 1 = a \quad \uparrow \text{inf}$$

III "distr" p₁ de "+"

$$\text{Distr : } (a+b)c = ac+bc \quad (\text{fermé})$$

IV Incl Boole $\forall x \in A \quad x^2 = x$

$$x^2 = x \cdot x = x \wedge x = x$$

b) Stim $(A, +, \cdot, 0, 1)$ imel Boole

Utem so dem $\mathcal{B}(A, \vee, \wedge, 0, 1, ')$ lattice Boole

I $(A, \vee, \wedge)$ lattice

Associativitt fr $\vee$: $(a \vee b) \vee c = a \vee (b \vee c)$
(Th Stone 6)

$$\begin{aligned}(a \vee b) \vee c &= (a + b + ab) \vee c = a + b + ab + c + (a + b + ab) \cdot c \\&= a + b + ab + c + ac + bc + abc \\&= a + b + c + ab + ac + bc + abc\end{aligned}$$

$$\begin{aligned}a \vee (b \vee c) &= a \vee (b + c + bc) = a + b + c + bc + a(b + c + bc) \\&= a + b + c + ab + ac + bc + abc\end{aligned}$$

Commutativitt fr $\vee$:

$$a \vee b \stackrel{?}{=} b \vee a$$

$$a \vee b = a + b + ab$$

$$b \vee a = b + a + ba$$

Absorbtie: $a \vee (a \wedge b) \stackrel{?}{=} a$

$$\begin{aligned}a \vee (a \wedge b) &= a \vee (a \cdot b) = a + ab + a \cdot a \cdot b = a + ab + a^2 b \\&\stackrel{a^2=a}{=} a + ab + ab \stackrel{11=a}{=} a\end{aligned}$$

Absorbtie fr $\wedge$: $a \wedge (a \vee b) = a$

$$\begin{aligned}a \wedge (a \vee b) &= a \wedge (a + b + ab) = a(a + b + ab) = a^2 + ab + a^2 b \\&= a + ab + ab = a\end{aligned}$$

$$a \wedge b = a \cdot b$$

distn

} $\Rightarrow$ "1" over si Com

II $(A, \cup, \cap)$ lattice distributive ternary

Ex: For M a multiset

$(\mathcal{P}(M), \cup, \cap, \emptyset, M, \cdot)$ lattice Boolean

$\Downarrow$ Th Stone

$(\mathcal{P}(M), \Delta, \cap, \emptyset, M)$ incl Boolean

$$A+B \stackrel{\text{Th Stone}}{=} (A \cap B) \cup (B \cap A) = (A \cap B) \cup (B \cap A) = A \Delta B !$$

$$A \cdot B = A \cap B$$

(108) $A, B, C \in \mathcal{P}(U)$ data for $x \in \mathcal{P}(U)$ measurement

$$\begin{aligned} \text{a) } \begin{cases} A \cap x = B \\ A \cup x = C \end{cases} &\stackrel{\text{Th Stone}}{\Leftrightarrow} \begin{cases} A \cdot x = B \\ A+x+A \cdot x = C \end{cases} \quad \text{①} \\ &A+x+A \cdot x+A \cdot x = B+C \\ &\quad \quad \quad \underbrace{}_{=0} \end{aligned}$$

$$A+x \mid A+x = B+C \quad \text{②}$$

$$x = A+B+C$$

$$\Rightarrow \boxed{x = A \Delta B \Delta C}$$

$$\text{b) } \begin{cases} A \setminus x = B \\ x \setminus A = C \end{cases} \Leftrightarrow \begin{cases} A \cap C \setminus x = B \\ x \cap C \setminus A = C \end{cases} \stackrel{\text{Th Stone}}{\Rightarrow} \begin{cases} A(1+x) = B \\ x(1+A) = C \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} A+A \cdot x = B \\ x+x \cdot A = C \end{cases} \quad \text{③}$$

$$A+x = B+C$$

$$x = A+B+C \Rightarrow x = A \Delta B \Delta C$$